

PROGRAM 9:-

AIM

To solve the Travelling Salesman Problem (TSP) by finding the shortest possible route that visits every city exactly once and returns to the starting city.

ALGORITHM

1. Start the program.
2. Create a list of cities and their distances.
3. Select a starting city.
4. Generate all possible routes between the cities.
5. Calculate the total distance for each route.
6. Compare all the calculated distances.
7. Select the route with the minimum total distance.
8. Display the shortest route and its total distance.
9. Stop the program.

PROGRAM

```
from itertools import permutations

cities = ['A', 'B', 'C', 'D']

distance = {
    'A': {'B': 10, 'C': 15, 'D': 20},
    'B': {'A': 10, 'C': 35, 'D': 25},
    'C': {'A': 15, 'B': 35, 'D': 30},
    'D': {'A': 20, 'B': 25, 'C': 30}
}

start = 'A'

others = [city for city in cities if city != start]

min_distance = float('inf')

best_route = None

for route in permutations(others):
    path = [start] + list(route) + [start]
```

```
total = 0

for i in range(len(path) - 1):
    total += distance[path[i]][path[i + 1]]
if total < min_distance:
    min_distance = total
    best_route = path

print("Shortest Route:", " -> ".join(best_route))
print("Minimum Distance:", min_distance)
```

OUTPUT

Shortest Route: A -> B -> D -> C -> A

Minimum Distance: 80

RESULT

The Travelling Salesman Problem was successfully solved by finding the shortest route that visits all the cities exactly once and returns to the starting city.